2025 28th International Conference on Computer and Information Technology (ICCIT)
19–21 December 2025, Cox's Bazar, Bangladesh

MalariaNet-XAI: An Explainable Deep Ensemble Framework for Accurate Malaria Detection

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2025 28th International Conference on Computer and Information Technology (ICCIT) | 979-8-3315-7867-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCIT68739.2025.11491174

***Abstract*—Malaria continues to pose a major global health challenge, where timely and precise diagnosis plays a vital role in successful treatment. Although deep learning presents a powerful automated solution compared to the traditional manual microscopic process, issues such as the opaque decision-making of many models and their limited ability to generalize remain key obstacles for clinical deployment. To overcome these limitations, we propose MalariaNet-XAI, an innovative framework built on a soft-voting ensemble of four structurally diverse Convolutional Neural Networks: VGG16, ResNet50, DenseNet121, and MobileNetV3. By combining heterogeneous feature representations, this ensemble improves both predictive performance and model reliability. Experiments conducted on the widely used NIH Malaria Dataset, which includes 27,558 cell images, demonstrate that our model achieves superior results, obtaining 98.04% accuracy and an Area Under the Curve (AUC) of 0.9964, surpassing the performance of the individual models. Furthermore, we incorporate Local Interpretable Model-agnostic Explanations (LIME) to visually highlight the parasitic regions influencing predictions, thereby improving model interpretability and fostering clinical trust. The combination of high accuracy and interpretability positions MalariaNet-XAI as a reliable tool for computer-aided malaria diagnosis, with the potential to augment clinical workflows in resource-constrained settings.**

***Index Terms*—Malaria Detection, Ensemble Learning, Deep Learning, Explainable AI (XAI), Computer-Aided Diagnosis, Medical Image Analysis**

I. Introduction

Malaria ranks among the major causes of mortality worldwide [1]. Malaria spread by female *Anopheles* mosquitoes bearing *Plasmodium* parasites. About 30 of the approximately 400 *Anopheles* species are important carriers. During the egg, larva, pupa, and adult phases of the mosquito life cycle, during which malaria is spread, females need human blood to develop their eggs. The five species responsible for malaria infection are *Plasmodium falciparum*, *Plasmodium vivax*, *Plasmodium ovale*, *Plasmodium malariae*, and *Plasmodium knowlesi*. Among these, *P. falciparum* and *P. vivax* are considered the most dangerous. After transmission through a mosquito bite, the parasite can remain dormant for 10 to 15 days before invading red blood cells and triggering symptoms such as fever, chills, nausea, and vomiting. The World Health Organization reports that regions including the Americas, the Western Pacific, the Eastern Mediterranean, and South-East Asia are at particularly high risk for malaria [2]. The WHO report from 2022 states that 249 million individuals with malaria were recorded in 85 endemic nations. One year after the global COVID-19 pandemic, the estimated malaria cases rose to 241 million, reflecting an increase of 14 million compared to 2019 [3]. To identify whether a person is infected or not, microscopists analyse millions of blood slides annually for malaria parasites [1].

Traditional malaria detection techniques, such as microscopic slide analysis, are frequently flawed by human error. PCR is more sensitive, but it needs certain tools and chemicals. Other methods are less sensitive, expensive, and time-consuming, like cell culture and pathogen isolation [4]. Recently, deep learning-based computer-aided diagnosis has demonstrated significant promise in the classification of malaria, improving automation, speed, and accuracy while reducing the workload for medical practitioners. The efficiency of computer vision has also been proven in a range of medical imaging tasks [5]. These methods are mostly used as computer-aided diagnostic (CAD) systems, which increase diagnosis accuracy and offer quick support [6].

Although deep learning has greatly advanced medical image analysis, several challenges remain, including achieving consistent performance on large datasets, avoiding overfitting when data is limited, and delivering explanations that are trustworthy for clinicians. Our work addresses these issues by combining ensemble learning with strong interpretability techniques, leading to a more reliable and transparent malaria classification diagnosis method.

The following are this study's principal contributions:

- To improve robustness and generalizability, images were cropped, divided into training (70%), validation (10%) and testing (20%) sets, and Data augmented by random cropping, flipping, rotation and jittering of the colors.

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- Assembling ResNet50 [7], DenseNet121 [8], MobileNetV3 [9], and VGG16 [10] into a strong ensemble deep learning pipeline. Our ensemble achieved state-of-the-art accuracy and excelled all individual CNN baselines on the Kaggle malaria dataset (27,558 images, evenly divided between Infected and Uninfected classes), exhibiting strong generalisation and dependability.
- Using explainable AI methods (LIME) [11] to draw attention to key cellular areas in parasitised and uninfected blood sample pictures. This interpretability technique bridges the gap among deep learning efficiency and real-world medical usage by guaranteeing that predictions are reliable, clinically meaningful, and correct.

II. Related Works

A number of recent research have employed deep learning techniques to classify malaria parasites. Yang et al. [12] introduced the first smartphone-assisted malaria detection framework for thick blood smears, which employed a customized CNN for classification together with Iterative Global Minimum Screening (IGMS) for candidate selection. Evaluated on 1,819 images obtained from 150 patients, the system achieved an accuracy of 93.46%, a precision of 92.59%, and a specificity of 94.33%. Although the method is inexpensive and portable, species-level classification is not taken into consideration, and patch-level screening may overlook uncommon parasites. Future enhancements ought to focus on speed optimisation and multi-class detection. Kassim et al. [13] developed a two-stage framework called RBCNet, which integrates U-Net for clustering-based cell segmentation with Faster R-CNN to improve the detection of red blood cells. The method achieved over 97% accuracy in detecting on a dataset of approximately 200,000 labelled cells, outperforming CNN-only methods. Although it uses a lot of processing power and depends on large annotated datasets, it performs exceptionally well in dense and overlapping cells. Future studies should include parasite classification and increase adaptability across staining variations. Sukumarran et al. [14] presented a two-stage deep learning approach that combines DenseNet-121 to parasite classification with YOLOv4 for identifying cells infected with malaria. In detecting *P. falciparum*, *P. malariae*, *P. vivax*, and *P. ovale*, DenseNet-121 obtained 95.5% accuracy, whereas YOLOv4 scored 89% mAP on the source dataset and 90% on external validation. The technique effectively manages overlapping cells and lessens reliance on human segmentation, but it comes at a significant computational cost. The method should be expanded in future research to include novel species like *P. knowlesi* and stage classification. To facilitate malaria diagnosis, Goni et al. [15] proposed a CNN-based framework combined with a Double Hidden Layer Extreme Learning Machine (CNN-DELM). To increase small parasitic areas in red blood cell pictures, a preprocessing procedure known as a parasite inflator was employed. The model outperformed CNN and CNN-ELM baselines with an accuracy of 97.79% on the original dataset and 99.66% on a clean version of the 27,558-image Kaggle malaria dataset. Although the method is effective, its performance heavily relies on the quality of the preprocessing steps. Testing on a variety of datasets and enhancing clinical interpretability should be the main goals of future research. For sequence-based malaria image classification, Alonso Ramírez et al. [16] introduced a hybrid recurrent convolutional neural network (CRNN) that combines CNN feature extraction with LSTM layers. When evaluated on the NIH malaria dataset, this model surpassed standalone CNNs, achieving an accuracy of 97.4%. Although the model is more computationally demanding, the incorporation of temporal dependencies improved feature representation. The architecture should be modified in future work for real-time, lightweight applications in settings with constrained resources. Yebasse et al. [17] introduced a novel highlighting method to guide CNNs to focus on small, fuzzy parasite regions in malaria-affected red blood cells. Tested on the 19,000-image NIH malaria dataset, the method improved classification accuracy across four architectures (Model-S, Model-M, ResNet-50, and MobileNet). The accuracy of the top performance was 97.2%, higher than the accuracy of the baseline, which was 94.49%. The approach increases generalisation by reducing overfitting and acts as a sort of regularisation, despite being unique to this dataset and possibly requiring modification for broader clinical use. In subsequent work, regions of interest will be automatically highlighted, and the approach will be expanded to detection and segmentation tasks.

III. Methodology

The proposed automated malaria detection framework, MalariaNet-XAI, is depicted in Fig. 1. The pipeline is designed as an end-to-end system, beginning with extensive data preprocessing and augmentation to enhance the quality of the images for analysis. Next, a set of diverse pre-trained deep convolutional neural networks (CNNs) is fine-tuned to extract distinctive features from the cell images. The outputs of these individual models are then combined using a soft voting ensemble approach to generate a final, robust classification. To increase clinical interpretability and trust, Local Interpretable Model-agnostic Explanations (LIME) are used to highlight the cellular regions that most strongly influence the model's decisions.

A. Dataset and Preprocessing

We used the publicly available "Malaria Cell Images Dataset" from the NIH [18], which contains 27,558 Giemsa-stained thin blood smear images evenly split between 'Parasitized' and 'Uninfected' classes. The dataset was divided into training, validation, and test sets in a 70%, 10%, and 20% ratio using stratified sampling to preserve class balance. All images were resized to 224 $\times$ 224 pixels and normalized using the ImageNet mean ([0.485, 0.456, 0.406]) and standard deviation ([0.229, 0.224, 0.225]).

To enhance generalization and reduce overfitting, data augmentation was applied to the training set, including random resized cropping (80–100% of the original area), horizontal flipping (0.5 probability), rotations within $\pm$15 degrees, and

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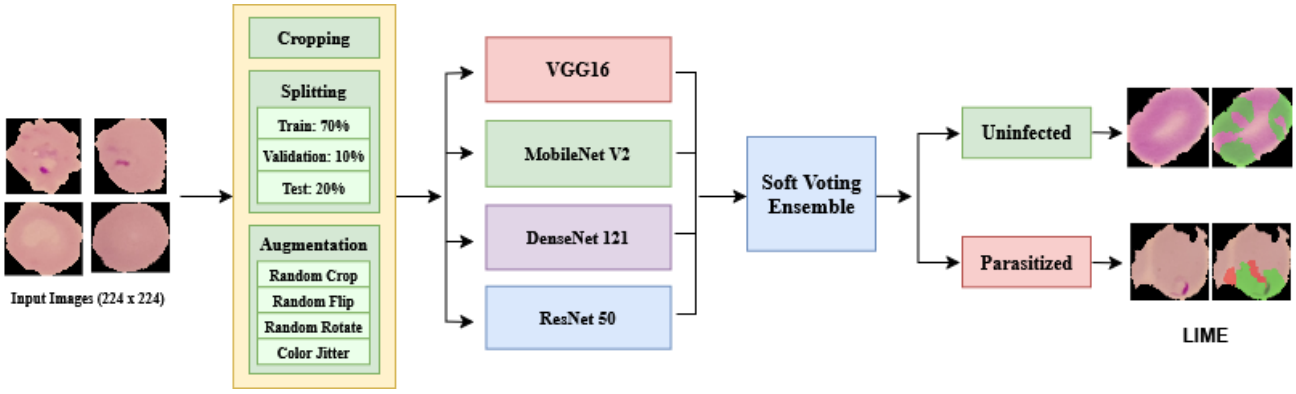


Fig. 1. The Proposed MalariaNet-XAI Framework

color jittering with brightness and contrast adjustments up to 20%, saturation up to 10%, and hue up to 5%.

B. MalariaNet-XAI Architecture and Ensemble Learning

Our classification framework is based on transfer learning, which utilizes knowledge from models previously trained on large-scale datasets. We selected four distinct and powerful CNN architectures as base learners for our ensemble. These models were chosen for their varied architectural philosophies, ensuring a diverse set of feature extractors. The base models include **VGG16** [10], featuring a straightforward deep architecture of stacked 3×3 convolutional layers. **ResNet50** [7], which uses residual connections to facilitate training of deeper networks by addressing vanishing gradients. **DenseNet121** [8], where each layer connects to all subsequent layers in a dense block to enhance feature reuse and information flow. and **MobileNetV3** [9], a lightweight model optimized for limited-resource environments, employing depthwise separable convolutions and squeeze-and-excite blocks.

For each pre-trained model, the final classification layer, which was originally designed for the 1000-class ImageNet challenge, was substituted with a new fully connected layer that culminates in a softmax output for our binary classification task. During the training phase, the entire network was fine-tuned. This allowed the weights of both the convolutional base and the new classifier head to be updated, thereby adapting the learned features to the specific domain of malaria cell morphology.

To synthesize the predictions from these individual models, we implemented a soft voting ensemble. In contrast to hard voting, which depends on a simple majority vote, soft voting aggregates the probabilistic outputs from each base model. This method takes into account the confidence of each model in its prediction, which often leads to superior performance and stability. For a given input image I, each base classifier m in our set of $M = 4$ models produces a probability distribution over the classes. The ensemble's final prediction, $\hat{c}_{\text{ens}}$, is obtained by averaging the class probabilities and choosing the class with the highest aggregated score, as expressed in (1):

$$\hat{c}_{\text{ens}} = \arg\max_i \left(\frac{1}{M} \sum_{m=1}^{M} \pi_m(i|I) \right) \tag{1}$$

where $\pi_m(i|I)$ represents the probability assigned to class i for image I by the m-th model. This ensemble strategy effectively reduces model variance and capitalizes on the diverse feature representations learned by each architecture, leading to a more robust and accurate diagnostic tool.

C. Explainability with LIME

A major obstacle to using deep learning in clinical settings is their "black-box" behavior. To address this, we incorporated an explainability module based on Local Interpretable Model-agnostic Explanations (LIME) [11], which provides insight into the model's decisions by creating a locally interpretable approximation for each prediction.

For any given image, LIME creates a set of perturbed instances by dividing the image into contiguous superpixels and selectively hiding subsets of them. These perturbed images are then passed through our trained ensemble model to observe the resulting changes in prediction probabilities. By fitting a sparse linear model to this data, LIME identifies the superpixels that had the most significant positive or negative impact on the predicted class probability. The resulting visualization highlights these critical regions on the original image. This allows a human expert to verify whether the model is focusing on clinically relevant features, such as the parasite within a red blood cell, or if it is relying on spurious correlations. This step is crucial for building trust and facilitating the translation of our automated system into real-world diagnostic workflows.

IV. Results and Discussion

A. Experimental Setup

All experiments were performed on a workstation featuring an NVIDIA Tesla V100 GPU with 16 GB of VRAM, an Intel Xeon CPU, and 32 GB of RAM. The deep learning models were implemented in Python 3.8 using the PyTorch framework. Model architectures and data transformations were sourced from torchvision, while performance metrics were

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computed using scikit-learn, and the interpretability maps were generated with the LIME library [11].

The models were trained with the AdamW optimizer [19], which separates weight decay from gradient updates to improve generalization. CrossEntropyLoss was used as the loss function for this binary classification task. A `ReduceLROnPlateau` scheduler dynamically reduced the learning rate by a factor of 0.2 if validation accuracy did not improve for two consecutive epochs. Early stopping was applied to halt training after five epochs without improvement in validation accuracy. Training efficiency was further enhanced using automatic mixed-precision. Key hyperparameters for all models are listed in Table I.

TABLE I
TRAINING HYPERPARAMETERS FOR ALL MODELS

Hyperparameter	Value
Optimizer	AdamW
Learning Rate	1×10^{-4}
Weight Decay	1×10^{-3}
Batch Size	32
Max Epochs	20
Loss Function	Cross-Entropy
LR Scheduler	ReduceLROnPlateau
Early Stopping Patience	5 epochs
Image Input Size	224×224 pixels

B. Model Convergence and Training Analysis

Fig. 2 displays the training and validation curves for VGG16, ResNet50, MobileNetV3, and DenseNet121. With progressively declining loss and rising accuracy over epochs, all models show stable convergence. MobileNetV3 reaches fast convergence in 7 epochs, but DenseNet121 converges steadily with good generalisation. VGG16 adapts well despite beginning with a little larger loss, while ResNet50 strikes a compromise between depth and stability, convergent about epoch 10. Early stopping confirmed dependable learning dynamics and decreased overfitting by ensuring that each model's optimal generalisation points were recorded.

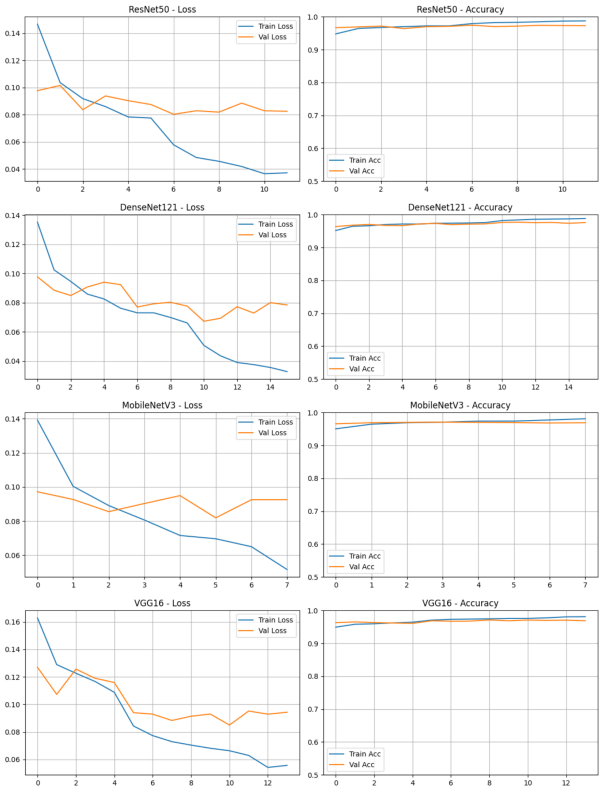


Fig. 2. Training and validation curves of DenseNet121, MobileNetV3, ResNet50, and VGG16 models

C. Confusion Matrix

The confusion matrix of the proposed ensemble model, shown in Fig. 3, illustrates its ability to accurately classify both parasitized and uninfected cells. The model accurately identified 1341 parasitised (True Positives) and 1361 uninfected (True Negatives) from a total of 2756 test images. While 17 uninfected samples were mistakenly projected to be parasitised (False Positives), only 37 parasitised sample were incorrectly classified as uninfected (False Negatives). This distribution demonstrates the model's outstanding specificity for uninfected cells and high sensitivity for identifying parasitised cells. The ensemble strategy's robustness and dependability for diagnosing malaria are confirmed by the minimal number of misclassifications.

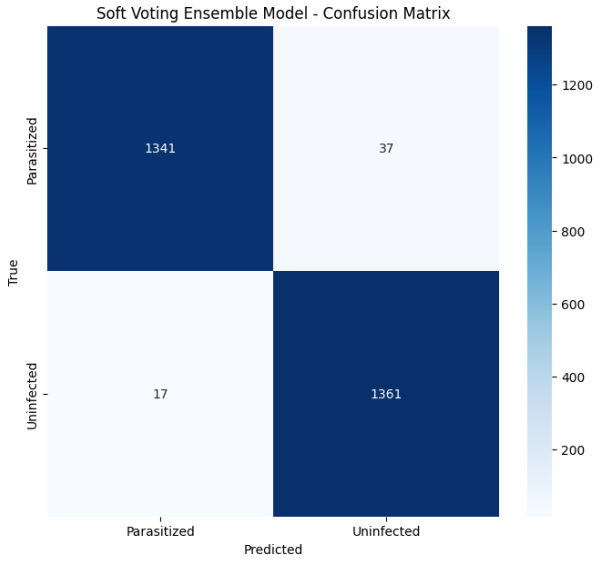


Fig. 3. Confusion Matrix for the proposed MalariaNet-XAI model

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D. AUC-ROC Curve

The ROC curve of the proposed ensemble model, shown in Fig. 4, depicts the relationship between the True Positive Rate and False Positive Rate across different thresholds. The model demonstrated exceptional discriminative performance between parasitised and uninfected red blood cells with an AUC value of **0.9964**, which is very close to 1. Such a high AUC indicates both great specificity in accurately recognising uninfected cells and strong sensitivity in detecting samples infected with malaria. This validates the suggested ensemble approach's applicability for managing binary classification problems in malaria detection by showcasing its resilience and dependability.

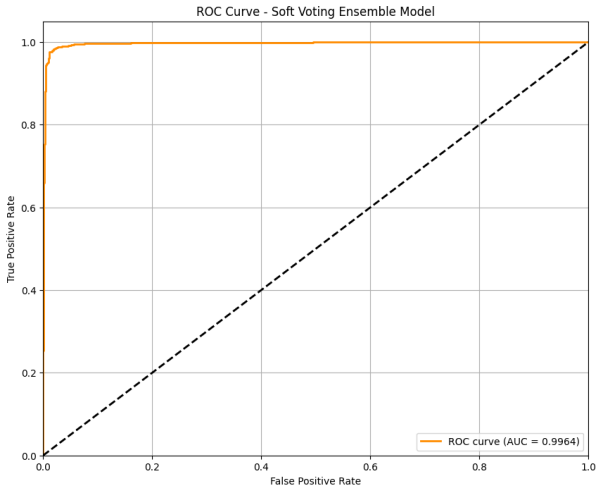


Fig. 4. ROC curve for the proposed MalariaNet-XAI model

E. Result Analysis

Table II presents a comparison of the proposed ensemble model against the baseline models.

TABLE II
PERFORMANCE COMPARISON OF THE ENSEMBLE AND INDIVIDUAL MODELS

Model	Accuracy	Precision	Recall	F1-score
MobileNetV3	0.9706	0.9709	0.9706	0.9706
ResNet50	0.9793	0.9793	0.9793	0.9793
VGG16	0.9728	0.9731	0.9728	0.9728
DenseNet121	0.9793	0.9792	0.9793	0.9793
MalariaNet-XAI	**0.9804**	**0.9805**	**0.9804**	**0.9804**

The findings confirm the strong performance of the MalariaNet-XAI framework, attaining 98.04% accuracy and an AUC of 0.9964. The success of our approach can be attributed to the soft voting ensemble strategy. By aggregating the probabilistic predictions from architecturally diverse models (VGG16, ResNet50, DenseNet121, and MobileNetV3), the ensemble effectively mitigates the biases and variances of individual learners. MalariaNet-XAI outperforms all individual CNN baselines, as demonstrated in Table II, attaining the best accuracy (98.04%) along with consistently better precision, recall, and F1-score. This illustrates how the ensemble technique produces a more dependable diagnostic framework by ensuring balanced performance across several evaluation parameters in addition to increasing classification accuracy. Furthermore, Table III illustrates MalariaNet-XAI's comparative benefit over earlier malaria categorisation techniques. Our system reached 98.04%, which is a significant improvement above previous methods like Ramírez et al. [16] and Goni et al. [15], which achieved accuracies of 97.4% and 97.79%, respectively. This improvement in performance highlights how well various CNN designs may be integrated through soft voting, improving generalisation and avoiding the drawbacks of individual models.

F. Explainability Analysis

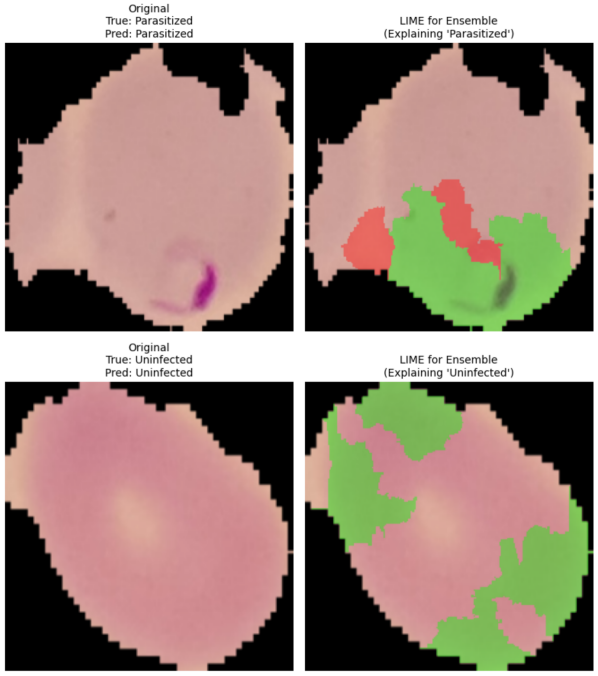


Fig. 5. LIME explanation for parasitized (top) and uninfected (bottom).

LIME explanations for the suggested ensemble model are displayed in Fig. 5. The model shows irregular, localised areas in the red blood cell in the parasitised case (top row), which match the unique forms of malaria parasites. The model has learnt to link irregular, clustered shapes with infection, as evidenced by these highlighted parasite-like structures. Conversely, the model highlights homogenous, wide, and smooth cell areas for the uninfected instance (bottom row). The contrast between the regular, rounding morphology of normal cells and parasite-specific abnormalities shows that the ensemble model relies its predictions on significant morphological cues. The difference between parasite-specific defects and normal cells' regular, rounded morphology demonstrates how the ensemble model depends on important morphological cues to make its predictions.

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TABLE III
COMPARING PERFORMANCE WITH PREVIOUS WORK

Author	Year	Method	Dataset	Accuracy
Yang et al. [12]	2020	Smartphone CNN + IGMS	Thick smear (1819 images, 150 patients)	93.46%
Kassim et al. [13]	2021	RBCNet (U-Net + Faster R-CNN)	∼200k cells (Bangladesh)	97.0%
Sukumarran et al. [14]	2024	YOLOv4 + DenseNet121 (Two-stage)	Multi-species	95.5%
Goni et al. [15]	2023	CNN + Double Hidden Layer ELM + Parasite Inflator	Five species dataset	97.79%
Ramírez et al. [16]	2022	CNN + LSTM (CRNN)	NIH Malaria Dataset	97.4%
Proposed MalariaNet-XAI	**2025**	**(MobileNetV3+DenseNet121+VGG16+ResNet50)**	**NIH Malaria Dataset**	**98.04%**

V. CONCLUSION

This study presents MalariaNet-XAI, a novel deep learning framework designed for automated malaria classification using thin blood smear images. By employing a soft voting ensemble of four distinct pre-trained CNN architectures, VGG16, ResNet50, DenseNet121, and MobileNetV3, our approach harnesses diverse feature representations to achieve exceptional diagnostic performance. The proposed ensemble achieved state-of-the-art performance on the benchmark NIH Malaria Dataset, reaching 98.04% accuracy and an AUC of 0.9964, surpassing all individual baseline models. Moreover, the use of LIME for explainability verified that the predictions rely on clinically relevant morphological features, helping to meet the essential requirement of transparency in medical AI. MalariaNet-XAI represents a significant step towards developing reliable and interpretable computer-aided diagnostic tools, offering the potential to support healthcare professionals in resource-limited settings by providing rapid and accurate malaria screening. In future work, we plan to explore the integration of Vision-Language Models (VLMs) to enhance multimodal reasoning, enable richer clinical context understanding, and further improve the interpretability and adaptability of malaria diagnostic systems.

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